CYBERSRS

Email Security Gateway Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links. The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages. The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events. The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

FR-001 - protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links

Statement: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Rationale: Source obligation E-001_O01.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: system designs a secure email security gateway

When: the system initiates protection against phishing, malicious attachments, spoofed messages, and suspicious links

Then: The system initiates protection against phishing, malicious attachments, spoofed messages, and suspicious links, and external test capture confirms the resulting response.

Method: test

Observable evidence: External test capture of the resulting response

FR-002 - inspect incoming and outgoing email traffic

Statement: The system shall inspect incoming and outgoing email traffic.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system inspects incoming and outgoing email traffic

When: incoming and outgoing email traffic is processed

Then: the system returns a verified inspection record of all messages

Method: inspection

Observable evidence: External test capture of the resulting state

FR-003 - detect suspicious messages

Statement: The system shall detect suspicious messages.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system detects suspicious messages in email traffic

When: a message exhibits known malicious patterns or behaviors

Then: the system identifies and flags the message as suspicious

Method: analysis

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-c2722a8ac444 is TBD.

FR-004 - quarantine potentially harmful emails

Statement: The system shall quarantine potentially harmful emails.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system quarantines potentially harmful emails

When: a message is identified as potentially harmful

Then: the email is held in quarantine with no specified storage mechanism

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-005 - generate security alerts

Statement: The system shall generate security alerts.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: the system generates security alerts

When: a suspicious or harmful message is detected

Then: The system generates a security alert upon detecting a suspicious or harmful message; the destination of the alert is undecided.

Method: demonstration

Observable evidence: External test capture of the resulting response

FR-006 - allow authorized administrators to review quarantined messages

Statement: The system shall allow authorized administrators to review quarantined messages.

Rationale: Source obligation E-002_O05.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: authorized administrators review quarantined messages

When: a message is in quarantine and accessible via admin interface

Then: administrators view the content and metadata of the quarantined message

Method: inspection

Observable evidence: External test capture of the inspection record

FR-007 - allow authorized administrators to release quarantined messages

Statement: The system shall allow authorized administrators to release quarantined messages.

Rationale: Source obligation E-002_O06.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: authorized administrators release quarantined messages

When: an administrator confirms a quarantined message is safe

Then: the quarantined message is released upon administrator confirmation

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-008 - maintain audit logs of filtering decisions and administrator actions

Statement: The system shall maintain audit logs of filtering decisions and administrator actions.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business; auditability; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: the system maintains audit logs of filtering decisions and administrator actions

When: an administrator performs a filtering decision or admin action

Then: audit logs are maintained and captured for each filtering decision and administrator action performed

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

FR-009 - integrate with the university identity system

Statement: The system shall integrate with the university identity system.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business; integration

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: the system integrates with the university identity system

When: an identity system user attempts to access email services

Then: the system authenticates the user using credentials from the university identity system

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-010 - provide a dashboard for monitoring email security events

Statement: The system shall provide a dashboard for monitoring email security events.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: the system provides a dashboard for monitoring email security events

When: an administrator accesses the dashboard interface

Then: the dashboard is accessible and displays email security event data

Method: inspection

Observable evidence: External test capture of the inspection record

3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-001_O02.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability

When: An analyst inspects or reviews that acceptance plan

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-002 - be secure for university-wide deployment

Statement: The system shall be secure for university-wide deployment.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be secure for university-wide deployment.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-003 - be reliable for university-wide deployment

Statement: The system shall be reliable for university-wide deployment.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be reliable for university-wide deployment.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-819bb986f419 is TBD.

NFR-004 - be scalable for university-wide deployment

Statement: The system shall be scalable for university-wide deployment.

Rationale: Source obligation E-004_O03.

Priority: should

Dimensions: performance

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be scalable for university-wide deployment.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-543394346815 is TBD.

NFR-005 - be manageable for university-wide deployment

Statement: The system shall be manageable for university-wide deployment.

Rationale: Source obligation E-004_O04.

Priority: should

Dimensions: usability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be manageable for university-wide deployment.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-5268b96e41b0 is TBD.

Related requirement ids: FR-001

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; NFR-001; NFR-002

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-008; FR-009; FR-010

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

7. Compliance and policy applicability matrix

Open decisions: Decision Q-c2d66755e6d0 is TBD.; Decision Q-c2722a8ac444 is TBD.; Decision Q-0b9232a4f753 is TBD.

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: authorized administrators

Detail: Proposed logical endpoint for the source-stated actor: authorized administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-006; FR-007

Component name: Actor: authorized administrators

Responsibility: Represent the source-stated actor: authorized administrators.

Email Security Engine

Detail: Proposed logical owner for FR-001, FR-002, FR-003, FR-004, FR-005, NFR-002.

Status: proposed

Evidence ids: E-001; E-002; E-004

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; NFR-002

Component name: Email Security Engine

Responsibility: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links. The system shall inspect incoming and outgoing email traffic. The system shall detect suspicious messages. The system shall quarantine potentially harmful emails. The system shall generate security alerts. The system shall be secure for university-wide deployment.

Administrator Console

Detail: Proposed logical owner for FR-006, FR-007, NFR-001.

Status: proposed

Evidence ids: E-001; E-002

Requirement ids: FR-006; FR-007; NFR-001

Component name: Administrator Console

Responsibility: The system shall allow authorized administrators to review quarantined messages. The system shall allow authorized administrators to release quarantined messages. The system shall be secure.

Audit and Logging Service

Detail: Proposed logical owner for FR-008, NFR-003.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-008; NFR-003

Component name: Audit and Logging Service

Responsibility: The system shall maintain audit logs of filtering decisions and administrator actions. The system shall be reliable for university-wide deployment.

University Identity Integration Module

Detail: Proposed logical owner for FR-009, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-009; NFR-004

Component name: University Identity Integration Module

Responsibility: The system shall integrate with the university identity system. The system shall be scalable for university-wide deployment.

Security Dashboard

Detail: Proposed logical owner for FR-010, NFR-005.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-010; NFR-005

Component name: Security Dashboard

Responsibility: The system shall provide a dashboard for monitoring email security events. The system shall be manageable for university-wide deployment.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-006, FR-007.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-006; FR-007

From component: Actor: authorized administrators

To component: Administrator Console

Data or action: The system shall allow authorized administrators to review quarantined messages. The system shall allow authorized administrators to release quarantined messages.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-006, FR-007.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-006; FR-007

Untrusted component: Actor: authorized administrators

Trusted component: Administrator Console

Enforcement: The system shall allow authorized administrators to review quarantined messages. The system shall allow authorized administrators to release quarantined messages.

9. Threats

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: authorized administrators

Goal: The system shall allow authorized administrators to review quarantined messages.

Story: As authorized administrators, I need the following behavior: The system shall allow authorized administrators to review quarantined messages.

Preconditions

1. authorized administrators review quarantined messages

Steps

1. a message is in quarantine and accessible via admin interface

Success outcome: administrators view the content and metadata of the quarantined message

Requirement ids: FR-006

Evidence ids: E-002

Id: UC-002

Actor: authorized administrators

Goal: The system shall allow authorized administrators to release quarantined messages.

Story: As authorized administrators, I need the following behavior: The system shall allow authorized administrators to release quarantined messages.

Preconditions

1. authorized administrators release quarantined messages

Steps

1. an administrator confirms a quarantined message is safe

Success outcome: the quarantined message is released upon administrator confirmation

Requirement ids: FR-007

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: system designs a secure email security gateway
2. When: the system initiates protection against phishing, malicious attachments, spoofed messages, and suspicious links

Expected result: Expected: The system initiates protection against phishing, malicious attachments, spoofed messages, and suspicious links, and external test capture confirms the resulting response.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: the system inspects incoming and outgoing email traffic
2. When: incoming and outgoing email traffic is processed

Expected result: Expected: the system returns a verified inspection record of all messages

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: the system detects suspicious messages in email traffic
2. When: a message exhibits known malicious patterns or behaviors

Expected result: Expected: the system identifies and flags the message as suspicious

Evidence to record: Evidence: External test capture of the resulting response

Test type: analysis

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: the system quarantines potentially harmful emails
2. When: a message is identified as potentially harmful

Expected result: Expected: the email is held in quarantine with no specified storage mechanism

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: the system generates security alerts
2. When: a suspicious or harmful message is detected

Expected result: Expected: The system generates a security alert upon detecting a suspicious or harmful message; the destination of the alert is undecided.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-006

Requirement ids: FR-006

Procedure

1. Given: authorized administrators review quarantined messages
2. When: a message is in quarantine and accessible via admin interface

Expected result: Expected: administrators view the content and metadata of the quarantined message

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-007

Requirement ids: FR-007

Procedure

1. Given: authorized administrators release quarantined messages
2. When: an administrator confirms a quarantined message is safe

Expected result: Expected: the quarantined message is released upon administrator confirmation

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-008

Requirement ids: FR-008

Procedure

1. Given: the system maintains audit logs of filtering decisions and administrator actions
2. When: an administrator performs a filtering decision or admin action

Expected result: Expected: audit logs are maintained and captured for each filtering decision and administrator action performed

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-009

Requirement ids: FR-009

Procedure

1. Given: the system integrates with the university identity system
2. When: an identity system user attempts to access email services

Expected result: Expected: the system authenticates the user using credentials from the university identity system

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-010

Requirement ids: FR-010

Procedure

1. Given: the system provides a dashboard for monitoring email security events
2. When: an administrator accesses the dashboard interface

Expected result: Expected: the dashboard is accessible and displays email security event data

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-011

Requirement ids: NFR-001

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability
2. When: An analyst inspects or reviews that acceptance plan

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-012

Requirement ids: NFR-002

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be secure for university-wide deployment.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-013

Requirement ids: NFR-003

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be reliable for university-wide deployment.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-014

Requirement ids: NFR-004

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be scalable for university-wide deployment.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-015

Requirement ids: NFR-005

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be manageable for university-wide deployment.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

14. SDLC recommendation and workflow

Selected model: Agile

Justification: No differentiating input evidence is provided to select among Agile, DevSecOps, Spiral, V-Model, Waterfall, Hybrid, or Incremental models. All candidates receive equal suitability scores of 40 out of 100. The decision remains provisional due to unknown factors including budget, delivery, failure consequences, feedback, legacy, regulation, schedule, stability, uncertainty, verification. Without explicit evidence on these factors, no model can be ranked as superior.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.; FR-002: The system shall inspect incoming and outgoing email traffic.; FR-003: The system shall detect suspicious messages.; FR-004: The system shall quarantine potentially harmful emails.; FR-005: The system shall generate security alerts.; FR-006: The system shall allow authorized administrators to review quarantined messages.; FR-007: The system shall allow authorized administrators to release quarantined messages.; FR-008: The system shall maintain audit logs of filtering decisions and administrator actions.; FR-009: The system shall integrate with the university identity system.; FR-010: The system shall provide a dashboard for monitoring email security events.
2. Review and baseline the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall be secure for university-wide deployment.; NFR-003: The system shall be reliable for university-wide deployment.; NFR-004: The system shall be scalable for university-wide deployment.; NFR-005: The system shall be manageable for university-wide deployment.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-004: The system shall quarantine potentially harmful emails.
2. Review implementation and acceptance evidence against FR-005: The system shall generate security alerts.
3. Review implementation and acceptance evidence against FR-006: The system shall allow authorized administrators to review quarantined messages.
4. Review implementation and acceptance evidence against FR-007: The system shall allow authorized administrators to release quarantined messages.
5. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
6. Review implementation and acceptance evidence against NFR-002: The system shall be secure for university-wide deployment.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall protect students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.; FR-002: The system shall inspect incoming and outgoing email traffic.; FR-003: The system shall detect suspicious messages.; FR-004: The system shall quarantine potentially harmful emails.; FR-005: The system shall generate security alerts.; FR-006: The system shall allow authorized administrators to review quarantined messages.; FR-007: The system shall allow authorized administrators to release quarantined messages.; FR-008: The system shall maintain audit logs of filtering decisions and administrator actions.; FR-009: The system shall integrate with the university identity system.; FR-010: The system shall provide a dashboard for monitoring email security events.
3. Implement the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall be secure for university-wide deployment.; NFR-003: The system shall be reliable for university-wide deployment.; NFR-004: The system shall be scalable for university-wide deployment.; NFR-005: The system shall be manageable for university-wide deployment.
4. In an authorized isolated test environment, execute TEST-001: Given: system designs a secure email security gateway When: the system initiates protection against phishing, malicious attachments, spoofed messages, and suspicious links Expected: The system initiates protection against phishing, malicious attachments, spoofed messages, and suspicious links, and external test capture confirms the resulting response.
5. In an authorized isolated test environment, execute TEST-002: Given: the system inspects incoming and outgoing email traffic When: incoming and outgoing email traffic is processed Expected: the system returns a verified inspection record of all messages
6. In an authorized isolated test environment, execute TEST-003: Given: the system detects suspicious messages in email traffic When: a message exhibits known malicious patterns or behaviors Expected: the system identifies and flags the message as suspicious
7. In an authorized isolated test environment, execute TEST-004: Given: the system quarantines potentially harmful emails When: a message is identified as potentially harmful Expected: the email is held in quarantine with no specified storage mechanism
8. In an authorized isolated test environment, execute TEST-005: Given: the system generates security alerts When: a suspicious or harmful message is detected Expected: The system generates a security alert upon detecting a suspicious or harmful message; the destination of the alert is undecided.
9. In an authorized isolated test environment, execute TEST-006: Given: authorized administrators review quarantined messages When: a message is in quarantine and accessible via admin interface Expected: administrators view the content and metadata of the quarantined message
10. In an authorized isolated test environment, execute TEST-007: Given: authorized administrators release quarantined messages When: an administrator confirms a quarantined message is safe Expected: the quarantined message is released upon administrator confirmation
11. In an authorized isolated test environment, execute TEST-008: Given: the system maintains audit logs of filtering decisions and administrator actions When: an administrator performs a filtering decision or admin action Expected: audit logs are maintained and captured for each filtering decision and administrator action performed
12. In an authorized isolated test environment, execute TEST-009: Given: the system integrates with the university identity system When: an identity system user attempts to access email services

Expected: the system authenticates the user using credentials from the university identity system

13. In an authorized isolated test environment, execute TEST-010: Given: the system provides a dashboard for monitoring email security events When: an administrator accesses the dashboard interface Expected: the dashboard is accessible and displays email security event data
14. In an authorized isolated test environment, execute TEST-011: Given: A draft acceptance plan for this quality goal: the system shall be secure against known attack vectors and maintain confidentiality, integrity, and availability When: An analyst inspects or reviews that acceptance plan Expected: The plan's measurable targets remain TBD or unconfirmed
15. In an authorized isolated test environment, execute TEST-012: Given: A draft acceptance plan for this quality goal: the system shall be secure for university-wide deployment. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
16. In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for this quality goal: the system shall be reliable for university-wide deployment. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
17. In an authorized isolated test environment, execute TEST-014: Given: A draft acceptance plan for this quality goal: the system shall be scalable for university-wide deployment. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
18. In an authorized isolated test environment, execute TEST-015: Given: A draft acceptance plan for this quality goal: the system shall be manageable for university-wide deployment. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
19. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
20. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-004: The system shall quarantine potentially harmful emails.
2. Review implementation and acceptance evidence against FR-005: The system shall generate security alerts.
3. Review implementation and acceptance evidence against FR-006: The system shall allow authorized administrators to review quarantined messages.

4. Review implementation and acceptance evidence against FR-007: The system shall allow authorized administrators to release quarantined messages.
5. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
6. Review implementation and acceptance evidence against NFR-002: The system shall be secure for university-wide deployment.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-004: The system shall quarantine potentially harmful emails.
2. Review implementation and acceptance evidence against FR-005: The system shall generate security alerts.
3. Review implementation and acceptance evidence against FR-006: The system shall allow authorized administrators to review quarantined messages.
4. Review implementation and acceptance evidence against FR-007: The system shall allow authorized administrators to release quarantined messages.
5. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
6. Review implementation and acceptance evidence against NFR-002: The system shall be secure for university-wide deployment.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Ranked candidates

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Model: Hybrid

Score: 40

Reasons: No differentiating input evidence

Model: Incremental

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure email security gateway for a university that protects students, faculty, staff, and administrators from phishing, malicious attachments, spoofed messages, and suspicious links.

Id: E-002

Source: description

Quote: The system should inspect incoming and outgoing email traffic, detect suspicious messages, quarantine potentially harmful emails, generate security alerts, and allow authorized administrators to review and release quarantined messages.

Id: E-003

Source: description

Quote: The platform should maintain audit logs of filtering decisions and administrator actions, integrate with the university identity system, and provide a dashboard for monitoring email security events.

Id: E-004

Source: description

Quote: The solution should be secure, reliable, scalable, and manageable for university-wide deployment.

Source references

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.7453

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.71

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7076

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:14 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7071

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-92ffa375cd05a8f5:14 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.7113

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

kb-0150d5352d0ec304:15 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 15

Page or section: text chunk 16

Relevance score: 0.7097

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-92ffa375cd05a8f5:19 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 19

Page or section: text chunk 20

Relevance score: 0.7107

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.32

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-002	TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-002	TEST-003, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-004	E-002	TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-005	E-002	TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-006	E-002	UC-001, TEST-006, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-007	E-002	UC-002, TEST-007, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-008	E-003	TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-009	E-003	TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-010	E-003	TEST-010, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-001	E-001	TEST-011, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-002	E-004	TEST-012, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-003	E-004	TEST-013, Requirements and verification planning, Agile delivery cycle, Release and handoff
NFR-004	E-004	TEST-014, Requirements and verification planning, Agile delivery cycle, Release and handoff

Requirement	Evidence / sources	Stories / tests / SDLC
NFR-005	E-004	TEST-015, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.32